DeshiCart BD

Excel Dashboard Competition

COMPETITION RULEBOOK

Transform raw sales data into an accurate, interactive and decision-ready dashboard

Competition Information	Details
Organized by	PUST Data Science Club
Training series	Excel Unlocked
Competition window	13 days
Submission deadline	20 September 2026, 11:59 PM (BST)
Maximum score	110 points, including optional Power BI bonus

Accuracy first. Insight next. Design with purpose.

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Read before you begin

This rulebook is the official participant guide. Keep it open while working and complete the final checklist before uploading your submission.

Remember: don't treat this as just a competition—treat it as a learning opportunity. We believe best learning happens through real practice.

Approach each task as a real business problem. Understand every question, requirement, and challenge, then try to solve them step by step. If you complete the project with dedication and curiosity, you will gain practical knowledge and experience that even a thousand courses may not be able to teach you.

Focus on learning, solving, and improving. Winning will be a bonus.

1. Competition Overview

Participants receive a raw, uncleaned dataset of DeshiCart BD sales for the financial year 1 July 2025 to 30 June 2026. The goal is to convert the data into a reliable analytical dashboard that management could use.

Required workflow

Raw Data > Cleaning > Transformation > Analysis > KPIs > Dashboard > Insights

Everything required to complete the project was covered during the two Excel Unlocked training sessions. No additional advanced technique is required.

2. Business Scenario and Objective

DeshiCart BD is an online marketplace selling electronics, fashion, home and kitchen goods, beauty products, groceries, books and stationery across all eight divisions of Bangladesh.

The company has closed its financial year. Revenue has grown, but profit remains flat, and management cannot clearly identify which products, categories, regions or customers are genuinely profitable.

The Head of Business Operations needs a single-screen dashboard that answers:

- How did the business perform during the year, and how did performance change month by month?
- Which categories, products, regions and customers make money, and which do not?
- Is discounting improving sales or reducing profit?
- Where should management take action first?

Competition objective

Transform the raw dataset into a professional Excel dashboard. Correct numbers and useful findings matter more than decoration. A visually attractive dashboard built on incorrect calculations will score poorly.

3. Dataset Description

The Raw_Data sheet contains one row per order line. Revenue, Cost and Profit are not provided and must be calculated.

Field	Description
OrderID	Unique order reference
OrderDate	Order date within FY 2025-26
CustomerID	Unique customer reference
CustomerName	Customer name
CustomerType	New, Regular, Premium or Corporate
Division	One of Bangladesh's eight divisions
District	District within the division
City	City or town
ProductID	Unique product reference
ProductName	Product title
Category	Product category
SubCategory	Product sub-category
Quantity	Units ordered
UnitPrice	Selling price per unit in BDT
UnitCost	Cost per unit in BDT
Discount	Decimal discount; 0.10 means 10%
SalesChannel	Website, Mobile App, Facebook Page or Phone Order
PaymentMethod	bKash, Nagad, Rocket, Card or Cash on Delivery
ShippingMethod	Standard, Express, Same-Day or Pickup Point
OrderStatus	Delivered, Shipped, Pending, Cancelled or Returned

3.1 Official text values

Field	Approved values
Category	Electronics; Fashion; Home & Kitchen; Beauty & Personal Care; Grocery; Books & Stationery
Division	Dhaka; Chattogram; Khulna; Rajshahi; Sylhet; Barishal; Rangpur; Mymensingh
SubCategory	Mobile & Accessories; Computers; Audio; Home Appliances; Menswear; Womenswear; Footwear; Bags & Accessories; Cookware; Furniture; Home Decor; Bedding; Skincare; Haircare; Makeup; Fragrance; Staples; Beverages; Snacks; Cooking Essentials; Academic Books; Notebooks; Office Supplies; Art Supplies
OrderStatus	Delivered; Shipped; Pending; Cancelled; Returned
CustomerType	New; Regular; Premium; Corporate

4. Participant Tasks

1. Clean and standardise the raw data.
2. Create the calculated columns required for analysis.
3. Answer all 25 business questions in Section 6.
4. Calculate all eight mandatory KPIs in Section 7.
5. Build a dashboard that meets every requirement in Section 8.
6. Write 5-8 numerical and actionable business insights.
7. Package and submit the completed project according to Section 10.

5. Data Cleaning and Calculation Rules

The dataset contains realistic quality problems. Participants must identify them and follow the rules below so that results remain comparable.

Rule	Requirement
R1	Delete rows that are completely blank.
R2	Remove duplicate records. OrderID is unique; two genuine orders cannot share one OrderID.
R3	Delete any row where Quantity is zero or negative.
R4	Where City is missing, enter Unknown.

Rule	Requirement
R5	Where Discount is missing, treat it as 0.
R6	Recover a missing CustomerName from another row with the same CustomerID.
R7	Recover error text in UnitCost from another row with the same ProductID. Do not delete the row or leave the value blank.
R8	Standardise text values using the official data dictionary in Section 3.1.
R9	Convert every date to a real Excel date and every numeric value to a real number.
R10	Do not delete a repairable row. Only R1, R2 and R3 permit row removal.

5.1 Required calculated columns

Calculated field	Required formula
Revenue	Quantity x UnitPrice x (1 - Discount)
Cost	Quantity x UnitCost
Profit	Revenue - Cost

Important

The final number of clean rows is part of the evaluation. Clean carefully and do not remove valid, repairable records.

6. Required Business Analysis

6.1 Sales Performance

Q1. What is total sales revenue for the financial year?

Q2. How does revenue change month by month across the year?

Q3. Which month was strongest and which was weakest, and by how much?

6.2 Profitability

Q4. What is total profit and the overall profit margin percentage?

Q5. What are the profit and margin percentage of each category?

Q6. Which category earns the most revenue, and which earns the most profit? If they differ, explain the business meaning.

Q7. Which sub-categories are the least profitable?

6.3 Product Analysis

Q8. What are the top five products by revenue?

Q9. What are the top five products by profit? Are they the same as the Q8 list?

Q10. Where is DeshiCart losing money? Identify the products, categories or trading conditions producing negative profit and explain the cause.

6.4 Customer Analysis

Q11. How many distinct customers are there?

Q12. Who are the top 10 customers by revenue?

Q13. What percentage of total revenue comes from the top 10 customers?

Q14. Which CustomerType generates the most revenue, and which generates the most profit?

6.5 Regional Analysis

Q15. Show revenue and profit by division.

Q16. Which are the top five districts by revenue?

Q17. Which locations are underperforming and deserve management attention?

6.6 Order Analysis

Q18. What are the total number of orders and total quantity sold?

Q19. What is the average order value?

Q20. How do orders split by OrderStatus, and what is the cancellation rate?

Q21. Which sales channel and payment method are used most?

6.7 Discount Analysis

Q22. What is the average discount given?

Q23. Group orders into discount bands: 0%, 1-10%, 11-20%, and above 20%. Show revenue, profit and margin percentage for each band.

Q24. At what discount level does the business stop making money?

Q25. Which CustomerType receives the deepest discounts, and is that justified by the profit generated?

7. Mandatory KPIs

Every dashboard must display the following eight KPIs using exactly these definitions.

Code	KPI	Definition
K1	Total Sales	Sum of Revenue across all clean rows
K2	Total Profit	Sum of Profit across all clean rows
K3	Profit Margin %	Total Profit / Total Sales
K4	Total Orders	Number of rows in Clean_Data
K5	Total Quantity	Sum of Quantity
K6	Average Order Value	Total Sales / Total Orders
K7	Average Discount	Average of the Discount column
K8	Total Customers	Distinct count of CustomerID

7.1 Optional extra KPIs

- Net Sales excluding Cancelled and Returned orders
- Cancellation Rate
- Return Rate
- Average Units per Order
- Revenue per Customer
- Top Category Share %

Definition rule

If you use a different KPI definition, state it clearly on the dashboard. An unlabelled KPI that disagrees with the official definition will be marked incorrect.

8. Dashboard and Insight Requirements

8.1 Minimum dashboard requirements

- Dashboard title and the period covered by the data
- All eight mandatory KPIs displayed as KPI cards

- A time-based trend showing monthly performance
- Category and/or product analysis
- Regional analysis
- Profitability analysis, not only revenue
- Top and bottom performer analysis
- Charts appropriate to each questions
- At least one working slicer or filter connected to the visuals
- Clear labels, readable number formatting and professional layout
- A visible section containing business insights

There is no single correct layout. Aim for a clear, professional dashboard that fits on one screen.

8.2 Business insight requirements

Write 5-8 insights. Every insight must explain what happened, where it happened, how large the effect was using a number, and what management should do next.

Weak insight

"Sales increased." This contains no location, magnitude, explanation or action.

Strong insight

"Electronics contributes 69% of revenue but only 36% of profit because its margin is 5.8%, compared with 31.7% for Fashion. Shifting homepage promotion toward Fashion could improve profit without requiring additional orders."

Every insight must be supported by a number that appears in the participant's own Analysis sheet.

9. Workbook Structure

Required sheet	Purpose
Student_Details	Complete all mandatory participant information.
Instructions	Retain this provided sheet; do not edit or delete it.
Raw_Data	Retain the original raw data without modification.
Clean_Data	Duplicate Raw_Data, rename the copy, and clean/transform data here.
Analysis	Include supporting calculations, PivotTables and answers to all questions.

Required sheet	Purpose
Dashboard	Create the final interactive Excel dashboard here.

The six required sheet names must be spelled exactly as shown. Additional sheets are allowed when useful.

Acceptable additional names: Lookup, Calculations, PivotTables, Supporting_Analysis and Insights.

Do not use unclear names such as Sheet1, Sheet2, New, Final, Test or Copy of ....

10. Submission Rules and File Naming

Deadline

Submit within the 13-day competition window. Final deadline: 20 September 2026 at 11:59 PM (BST). Late submissions will not be accepted.

10.1 Required ZIP package

Place all required files in one folder, confirm that they open correctly, compress the folder into one ZIP file, and upload that ZIP through the official Google Form.

Status	File	Required filename pattern
Mandatory	Completed Excel workbook	Dashboard_Competition_StudentName.xlsx
Mandatory	Excel dashboard screenshot(s)	Excel_Dashboard_StudentName_01.png
Optional	Power BI report	Dashboard_Competition_StudentName.pbix
Optional	Power BI dashboard screenshot(s)	PowerBI_Dashboard_StudentName_01.png

If there is more than one screenshot, continue the numbering as 02, 03 and so on. Screenshots may be PNG or JPG, but PNG is preferred for clarity.

10.2 ZIP filename

Required ZIP name

Dashboard_Competition_StudentName.zip

Example: Dashboard_Competition_AbdulAlim.zip

- Use the same participant name entered in Student_Details and the Google Form.
- Do not use spaces or special characters in the participant name.
- The actual Excel workbook is mandatory; screenshots alone will not be accepted.
- Excel must be submitted as .xlsx, not .xls or .csv.
- Keep formulas live. Do not paste the dashboard into the workbook as a picture.
- Power BI is optional. If included, both the .pbix file and dashboard screenshot(s) must be placed inside the ZIP.

10.3 Submission channel

Upload the correctly named ZIP file through the official submission Google form:

Submission form

https://forms.gle/65MTnR1Q62G4Kpdq6

Only one ZIP file should be uploaded. Do not submit project files through WhatsApp, Messenger, email, screenshots, PDFs or separate Drive links unless the organisers announce a technical exception.

11. Judging and Scoring

Judging priority

ACCURACY > ANALYSIS > BUSINESS INSIGHTS > USABILITY > VISUAL DESIGN

Judges will verify that dashboard KPIs match the Analysis sheet, charts use the correct data, totals and percentages reconcile, and every insight is supported by a number in the workbook.

Evaluation area	Points
Data Cleaning and Preparation	15
Calculations and Accuracy	15
Business Analysis	15
KPI Selection and Accuracy	10
Visualization	15
Dashboard Design and Usability	15

Evaluation area	Points
Interactivity	5
Business Insights	10
Core score	100
Optional Power BI bonus	+10
Maximum possible score	110

12. Power BI Bonus

- The Excel workbook is mandatory. A Power BI-only submission is invalid and receives zero.
- Submitting both Excel and Power BI may earn up to 10 bonus points.
- The Power BI report must use the same dataset and must not contradict the Excel results.
- Contradictory calculations may reduce or remove bonus marks.
- The bonus considers data preparation, modelling, KPI accuracy, visuals, interactivity, usability, insights and consistency with Excel.

13. Academic Integrity and Important Rules

- Do not modify the Raw_Data sheet. Copy it to Clean_Data and work in the copy.
- Do not delete or rename Student_Details, Instructions or Raw_Data.
- Complete the competition individually unless the organisers explicitly announce otherwise.
- Participants may use Excel help, training notes and the student guide.
- Do not copy another participant's workbook. Identical or copied submissions receive zero.
- If a requirement is ambiguous, state the assumption clearly in the Analysis sheet. A stated assumption will be considered; a silent assumption may be marked incorrect.
- Submit before the official deadline. Technical problems reported after the deadline do not automatically qualify for an extension.

14. Final Submission Checklist

Check	Before uploading, confirm that...
[ ]	Student_Details is complete and the name matches all filenames.
[ ]	Instructions sheet is retained.
[ ]	Raw_Data is retained without modification.
[ ]	Clean_Data, Analysis and Dashboard sheets are present.
[ ]	All eight mandatory KPIs are present and correct.
[ ]	All 25 required business questions are answered.
[ ]	Five to eight business insights are included, each with a number and an action.
[ ]	The dashboard is readable, functional and includes at least one working slicer or filter.
[ ]	Required sheet names are spelled exactly.
[ ]	The actual Excel workbook opens without errors or missing links.
[ ]	Excel dashboard screenshot(s) are clear and included.
[ ]	Power BI file and screenshot(s) are included if claiming the bonus.
[ ]	Every file and the ZIP package follow the naming convention.
[ ]	The ZIP contains no unrelated files and has been uploaded through the official form.

Final reminder

Accuracy is the highest priority. Check every total, percentage, formula, chart source and insight before submitting. Keep a backup copy of the final ZIP package.

Good luck!